

EDUCATION

University of Illinois Chicago | *M.S. in Computer Science (Thesis Track)*

(Aug 2024 - May 2026)

Focus: Multimodal Summarization, Generative AI, LLM Systems

Vellore Institute of Technology, India | *Bachelor's in Computer Science*

(Aug 2019 - May 2023)

Focus: Artificial Intelligence, Machine Learning | GPA: 3.6/4.0

WORK EXPERIENCE

AI Intern - Carplai | RSNA (Nov 2025 - Dec 2025)

- Built and deployed multimodal medical-image retrieval workflows using **vision-language models**, improving diagnostic query relevance by **~28%**.
- Designed evaluation pipelines for LLM-based reasoning using prompt chaining and self-consistency to reduce hallucinations.
- Optimized inference latency for production-grade models by implementing batching, caching, and GPU-aware preprocessing, achieving a **35%** speedup.

Graduate Research Assistant - University of Illinois Chicago (Aug 2024 - Present)

Research focus: Multimodal Summarization, LLM reasoning, grounded generation, and evaluation.

- Built LLM inference pipelines in PyTorch and TensorRT, reducing latency by **~30%** and supporting large-scale evaluation of factual consistency in domain-specific generation.
- Implemented distributed inference workflows on AWS (EC2, Inferentia), focusing on batching, throughput, and monitoring.
- Conducting thesis research on multimodal summarization, focusing on grounding, evaluation, and hallucination reduction in domain-specific NLP.

Software Development Engineer I - BYJU'S (Jan 2024 - Jul 2024)

- Deployed backend compute workflows serving **1M+ daily users**, reducing system latency by **35%** and improving throughput by **22%**.
- Resolved **40+ runtime-critical defects**, improving memory stability and failure recovery in distributed environments.
- Collaborated across product & infra teams to improve **execution profiling, monitoring, and rollout safety**.

Data Scientist Intern - Extech Digital (May 2023 - Oct 2023)

- Released **10+ ML models** to production with **zero rollback**, overseeing deployment sequencing & validation workflows.
- Led analytical modeling informing pricing strategy changes driving **~30% revenue uplift**.
- Built automated monitoring dashboards to **track model drift** and execution behavior.

Research Assistant - VIT (Jan 2023 - May 2023)

- Developed ML-based DDoS detection model achieving **85%** accuracy on **10K+ network traces**, while reducing preprocessing time by **40%**.
- Benchmarked model performance under simulated network load; automated experiment orchestration via Python and shell scripts.
- Constructed reproducible experiments and benchmarks for network traffic classification.

PROJECTS

Speculative Decoding for LLM Inference Optimization - PyTorch · CUDA · TensorRT

- Integrated **speculative decoding** for LLaMA-3.1 and Gemma models, enabling **parallel forward prediction** to lower reasoning-step latency.
- Reduced iterative reasoning latency by **>40%** in agentic planning loops while preserving accuracy on GSM8K-style reasoning tasks.
- Benchmarked across **GSM8K, SVAMP, MultiArith**, maintaining reasoning accuracy while improving computational throughput.

Ethical RAG for Medical Information Retrieval - Python · Hugging Face · ChromaDB · AWS

- Built a hybrid retrieval system (BM25 + dense embeddings) with trustworthiness scoring, self-consistency validation, and confidence-based refusal to reduce hallucinations in clinical text generation.
- Processed 2,500+ PubMed passages into structured embeddings, improving domain-specific retrieval precision by **~25%** and achieving an AUC of **~0.89** in hallucination detection.
- Deployed a scalable RAG inference service on AWS with caching and batching optimization, supporting low-latency generation for real-world healthcare queries.

CHQ-Summ: Consumer Healthcare Question Summarization Dataset | Lead Author, submitted to Nature Scientific Data

- Contributed to dataset construction from Yahoo! Answers healthcare data, including annotation schema design and evaluation guidelines.
- Benchmarked **T5-Large, BART**, and **LLMs (LLaMA-3.1 8B, Gemma, Mistral 7B, DeepSeek-Coder)** achieving **ROUGE-L 49%** and **BERTScore 92%**.
- Released dataset + evaluation framework on GitHub: github.com/abasu9/CHQ-Summ-A-Dataset-for-Consumer-Healthcare-Question-Summarization

Friday: Privacy-First AI Meeting Assistant | MIT Frontiers GenAI Hackathon Finalist (Top 5) | Rust · Tauri · Next.js · FastAPI · SQLite · Whisper

- Developed a privacy-first, local-first AI meeting assistant for live transcription, meeting summaries, and editable export workflows, avoiding cloud upload of raw audio and transcripts by default.
- Engineered a cross-platform desktop stack with **Tauri/Rust, Next.js, FastAPI**, and **SQLite**, integrating local transcription pipelines and backend summarization endpoints for end-to-end meeting workflows.
- Implemented real-time transcript streaming, synchronized mic + system-audio processing, and editable/exportable meeting notes for professional productivity use cases.

TECHNICAL SKILLS

Programming: Python, C++, Java

Machine Learning / NLP: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers

Generative AI / LLM Systems: RAG, Hybrid Retrieval, Prompt Engineering, Self-Consistency, Speculative Decoding, LLM Evaluation

Inference & Optimization: CUDA, TensorRT, Batching, Caching

Cloud / Backend: AWS (EC2, S3, Inferentia), Docker, FastAPI, Flask

Tools / Systems: Git, Linux

Databases / Retrieval: PostgreSQL, MySQL, MongoDB, ChromaDB, BM25

ACHIEVEMENTS & CERTIFICATION

Finalist, Top 5 - MIT Frontiers GenAI Hackathon (in collaboration with Google DeepMind and Breakthrough Ventures)

Oracle Cloud Infrastructure - **Generative AI Professional (2025)**

Oracle Cloud Infrastructure - **AI Foundations Associate (2025)**

Basu, A., et al. "CHQ-Summ: A Dataset and Benchmark for Consumer Healthcare Question Summarization." Submitted to Nature Scientific Data, 2025.